Physical spectral weights in the local dS_3 crossing limit

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Working research note — 31 October 2026

Abstract

We insert the actual regulated creation amplitudes into the previously derived local scalar crossing limit. The representation-label change of variables has unit absolute Jacobian. After the physical normalization is included exactly once, the weights form a uniformly bounded C^2 family on a fixed compact transverse window. This proves a local, smoothly cut off physical pairing limit. It does not remove the cutoff or prove the full Bose free-observer limit. Numerical checks reuse the saved complete-kernel nodes and independently assemble their weights from the original vertices and four-body norms.

1 State, measure, and exact weight

Keep $\ell = 1$, $G_N = 0$, proper $SO^+(3,1)$, two distinguishable BD scalar observers with labels at least a fixed $s > 0$, KG matter $p, q > 0$, unit local coupling, and the half-period regulated creation blocks. Time is generated by $h(x) = \sqrt{1+x^2}$. The initial preparation here is conditional label $c = M$, defined as the limit of normalized narrow mass packets. It is not a fixed initial packet sent to large mass without changing its preparation.

Our earlier channel conventions [1, 2] give the coefficient of $U_{00}(b, d; a, q, c, p)$ in the crossed pairing as

$$A_M = \frac{g(a)}{g(c)} \sqrt{\widehat{R}_q(a,b)\widehat{R}_p(b,c)\widehat{R}_p(a,d)\widehat{R}_q(d,c)}, \tag{1}$$

$$\Phi_M(t) = \exp\{it[h(a) + h(c) - h(b) - h(d)]\}, \quad F_M = A_M \Phi_M, \tag{2}$$

$$g(x) = \frac{x e^{\pi[h(x)-x]}}{1 + e^{-2\pi x}}, \qquad \widehat{R}_p(x,y) = \frac{A - B}{8\pi(A+C)(B+C)}, \tag{3}$$

$$A = \cosh \pi(x+y), \quad B = \cosh \pi(x-y), \quad C = \cosh \pi p.$$

In these conventions the conditional integration measure is $\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}d$. Before conditioning it is $\pi_0(c)\,\mathrm{d}c\,\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}d$, where $\pi_0(c) = \rho(c)|f(c)|^2$, $\rho(c) = 2c^2/\pi$.

The same amplitude, before cancellation, is

$$A_M = \frac{\overline{E_{abq} E_{bcp}} E_{adp} E_{dcq} e^{\pi[h(a)-h(c)]}}{\rho(c)\sqrt{W_s W_t}}, \qquad W_s = \rho(b) K_{abq} K_{bcp}, \quad W_t = \rho(d) K_{adp} K_{dcq}.$$

The fixed spherical vertices are real positive in this convention. These equations already contain the observer and CG normalizations; multiplying another $\rho(b)\rho(d)$ into the integral would be incorrect.

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2 Coordinates and a uniform weight lemma

Set

$$c = M, \quad b = M + u, \quad a = M + u + v, \quad d = M + v + r.$$

Both $|\partial(a, b, c, d)/\partial(M, u, v, r)|$ and the conditional $|\partial(a, b, d)/\partial(u, v, r)|$ equal one. If one instead uses energies $\epsilon_x = h(x)$, then

$$\mathrm{d}a\,\mathrm{d}b\,\mathrm{d}d = \prod_{x=a,b,d} \frac{h(x)}{x}\,\mathrm{d}\epsilon_a\,\mathrm{d}\epsilon_b\,\mathrm{d}\epsilon_d.$$

The factor tends to one with $O(M^{-2})$ error on compact transfers, but it is not identically one. Conditioning in energy uses the transformed probability density $\pi_0(c)h(c)/c$.

Define $k_p(x) = [8\pi(\cosh \pi x + \cosh \pi p)]^{-1}$ and

$$F_\infty(u, v, r; t) = \sqrt{k_q(v)k_p(u)k_p(u - r)k_q(v + r)}\, e^{-itr}. \tag{4}$$

On any fixed compact set of real (u, v, r) and positive (p, q), uniformly for $|t| \le T$, the exact weights satisfy

$$F_M = \left(1 + \frac{u + v}{M}\right) F_\infty + O_{C_r^2,T}(M^{-2}). \tag{5}$$

All masses exceed s for sufficiently large M on this set. Here C_r^2 controls the function and its first two r derivatives; it is not an assertion for growing support or growing time.

Proof. The exact identity

$$\widehat{R}_p(x, y) = k_p(x - y)\left[1 - \frac{\cosh \pi(x - y) + \cosh \pi p}{\cosh \pi(x + y) + \cosh \pi p}\right]$$

has bracket $1 + O_{C_r^2}(e^{-2\pi M})$ for $x = M + O(1)$, $y = M + O(1)$. Derivatives of its numerator are bounded on the compact set; derivatives of the reciprocal denominator retain the exponential bound. The bracket is bounded away from zero for large M, so taking the positive square root preserves the estimate.

Write $\delta(x) = h(x) - x = 1/(h(x) + x)$. Its derivatives obey

$$|\delta'(x)| \le \frac{1}{2x^2}, \qquad \delta''(x) = (1 + x^2)^{-3/2} \le x^{-3} \quad (x > 0).$$

Consequently $h(a) + h(c) - h(b) - h(d) = -r + O_{C_r^2}(M^{-2})$. Also

$$\frac{g(M + u + v)}{g(M)} = \left(1 + \frac{u + v}{M}\right) e^{\pi[\delta(M+u+v)-\delta(M)]} \frac{1 + e^{-2\pi M}}{1 + e^{-2\pi(M+u+v)}} = 1 + \frac{u + v}{M} + O(M^{-2}).$$

This factor is independent of r; the derivative bounds for the phase, the product rule, and the positivity of the compact limiting weights prove (5). No differentiation of an unspecified remainder is used.

3 A local physical pairing theorem

Let $R > 0$ be fixed and let $\chi(u, v, r)$ be a fixed C^2 compactly supported cutoff, supported in $r \in [-R, R]$ and a compact transfer set on which

$$\max_j |\alpha_j| + R/2 \le 1/2, \qquad \alpha = \tfrac{1}{2}(v + q, p - u, -u - p, v - q). \tag{6}$$

This sufficient small-window restriction comes from the earlier kernel estimate; the weight lemma itself does not require it. Define the compact part of the actual conditional pairing,

$$I_M^\chi(t) = \int \chi(u, v, r) F_M(u, v, r; t) U_{00}(M; u, v, r)\, \mathrm{d}u\, \mathrm{d}v\, \mathrm{d}r.$$

The preceding local-kernel note [3] gives, uniformly on this set,

$$U_{00} = \frac{\sin(r \log M - \vartheta(u, v, r))}{\pi r} + O(M^{-1}), \qquad \vartheta(u, v, 0) = 0,$$

including the removable value at $r = 0$. Its Dirichlet estimate applies uniformly to fixed-support test families with bounded C_r^2 norms. The lemma supplies those bounds for χF_M, and hence

$$I_M^\chi(t) = \int \chi(u, v, 0) k_p(u) k_q(v)\, \mathrm{d}u\, \mathrm{d}v + O_{\chi,T}(1/\log M) + O_{\chi,T}(1/M). \tag{7}$$

The convergence is uniform on compact real times. To see the last integration explicitly, apply the one-dimensional estimate for each (u, v); the cutoff's support has finite area and the constants are uniform there. At $r = 0$, (5) gives $F_M = k_p(u)k_q(v) + O(M^{-1})$ uniformly. This completes the local argument.

This result treats a compact contribution to the original observable. The cutoff is an analysis device; we have not replaced the original preparation or internal spectrum by a filtered physical model. The leading time independence applies to the crossed phase (2). It does not say every Bose exchange matrix element has a time-independent limit.

For equal species, the exchange phase $\exp\{it[h(d) - h(b)]\}$ instead tends to $e^{it(v+r-u)}$. The same local argument then gives the integrand $\chi(u, v, 0)k_p(u)k_p(v)e^{it(v-u)}$. If a future global theorem permits removal of all cutoffs, these two expressions would give $L_p(0)^2$ and $|L_p(t)|^2$, respectively, with

$$L_p(t) = \frac{\sin(pt)}{4\pi \sinh(\pi p) \sinh t}, \qquad L_p(0) = \frac{p}{4\pi \sinh(\pi p)}.$$

They are the two different terms needed in the Bose Gram permanent. At present, replacing χ by one is unproved: the restricted domain (6) cannot exhaust the full transfer plane, and the original oscillatory pairing need not be absolutely integrable. A global argument must combine a larger compact-kernel theorem with the physical Hilbert norm tail bounds. In particular U_{00} being a contraction matrix kernel does not imply the pointwise bound $|U_{00}| \le 1$.

4 Numerical evidence and continuation

The new diagnostics use 85-digit arithmetic and preserve full complex values. Twelve points at $M = 4, 16, 64, 256$ compare (1) with the independent $E/W/\rho$ expression. The largest relative difference is about $2.95\,10^{-83}$, against the recorded empirical tolerance 10^{-65}. The fixed samples include unequal species and nonzero transfers; they do not prove uniformity.

Seventy-two derivative comparisons check analytic derivatives of (4) against numerical differentiation, for both crossed and exchange phases and orders $0, 1, 2$. The corresponding finite-M derivatives and M^2-scaled remainders in (5) are saved as observations, without imposing a convergence threshold. Their largest moduli at the four masses are approximately 0.000474, 0.000506, 0.000508, 0.000508.

For the physical local integral at $(u, v, p, q) = (0.1, -0.1, 0.4, 0.4)$, use $\chi_r = (1 - (r/R)^2)^3$ on $[-R, R]$, zero outside, $R = 0.25$. The following are ratios to the limiting value $k_p(u)k_q(v)$, using the 48-node rule:

t	$M = 64$	$M = 256$
0	0.470982	0.552526
1	$0.469502 - 0.001076i$	$0.550854 - 0.001179i$
2	$0.465087 - 0.002135i$	$0.545862 - 0.002340i$

These values are far from the limiting ratio one. In particular the small imaginary component need not decrease monotonically at these masses. The 24/48 rules reuse all 144 complete complex kernel nodes from [3]; they are not 144 new evaluations of U_{00}. Twelve reassemblies of the weighted integrals agree with the independent vertex representation. One exact symbolic Jacobian check completes the 97 checks. All propagated series tails exclude quadrature and rounding.

The implementation is `code/ds3_physical_local.py`; diagnostics and parameters are in `results/stage4/ds3-physical-local/diagnostics.json`. The driver refuses to overwrite existing results. Input/output hashes, dependency versions and the compilation review accompany this note. The proof is the compact estimate above, not the finite sample count. The next research task is to extend the compact kernel estimate beyond (6), then control the complement in physical Hilbert norm. A fixed initial packet, a moving threshold, and growing observation time each require separate arguments. Full Fock domains, finite-G_N dynamics and a complete microscopic holographic dual remain open.

References

[1] Alan Day and Chengheng, *dS_3 two-matter sector and noncrossed limit*, project note `ds3-two-matter`, 31 October 2026.

[2] Alan Day and Chengheng, *Physical crossed packets*, project note `ds3-crossed-packet`, 31 October 2026.

[3] Alan Day and Chengheng, *A uniform transverse limit of the dS_3 crossing kernel and its local action on test functions*, project note `ds3-uniform-local`, 31 October 2026.

[4] J. Liu, E. Perlmutter, V. Rosenhaus and D. Simmons-Duffin, arXiv:1808.00612v3, Eqs. (3.35)–(3.38), underlying crossing kernel.

[5] W. Groenevelt, arXiv:math/0306424v1, Proposition 4.7, precedent for the removable-pole Dirichlet argument.